



सीएसआईआर- केन्द्रीय खनन एवं ईंधन अनुसंधान संस्थान CSIR-Central Institute of Mining and Fuel Research

(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद / Council of Scientific & Industrial Research)

बरवा रोड, धनबाद - 826015, झारखण्ड, भारत / Barwa Road, Dhanbad - 826015, Jharkhand, India

ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF & EQUIPMENT SAFETY

परीक्षण प्रमाण पत्र / TEST CERTIFICATE



प्ररूप परीक्षण प्रमाण-पत्र/ TYPE EXAMINATION CERTIFICATE

प्रारूप सं./FORMAT NO.: (CIMFR: DQM: FLP02: F-01: REV-01)

(ज्वाला-सह एवं उपकरण सुरक्षा/Flameproof & Equipment Safety)

Certificate No.: CMF 23 INEx 0187

Dated: 12/07/2023

ULR No.: TC23000000210F

(Issue No. 0)

- आवेदक/ Applicant : M/s. SAI Industries,
Plot No. 4709, Plastic Zone, GIDC,
Dist. Valsad, Sarigam - 396155 (Gujarat).
- विनिर्माता/ Manufacturer : Same as Above
- उपकरण/ Equipment : Flameproof-Weatherproof (IP-66) Control Panel.
- नामित/ Designated by : Cat. No. : SAI/RCP/1058
- सुरक्षा का प्रकार /Type of Protection & EPL: Ex db Gas Group IIB, EPL Gb & Db.
- मार्किंग /Marking : (Ex db IIB T6 Gb) (Ex tb/tc IIIC T85°C Db/Dc)

उक्त प्रमाणपत्र को इस सत्यापन के रूप में जारी किया जाता है कि नमूना/ प्रलेखन का मूल्यांकन किया गया एवं यह पाया गया कि यह निम्न आईएस/ आईईसी मानक सूची के अनुरूप है।/ This certificate is issued as verification that a sample/documentation was assessed and found to comply with the IS/IEC Standard list below.

यह प्रमाण-पत्र केवल उपकरणों के डिजाइन तथा निर्माण से ही संबंधित है। इसमें विनिर्माण प्रक्रिया और उपयोग/प्रयोग से संबंधित अपेक्षा/ आवश्यकता शामिल नहीं है।/ This certificate relates to only design and construction of the equipment. It does not cover requirement related to manufacturing process and usage.

7. मानक/ Standards: इस प्रमाण-पत्र एवं संबंधित दस्तावेजों की अनुसूची में विनिर्दिष्ट किए गए उपकरण और इसके लिए स्वीकार्य कोई भी भिन्नता निम्नलिखित मानकों का अनुपालन करते हुए पाए गए।/ The equipment and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards:

- | | |
|----------------------|---|
| IS/IEC 60079-0:2017 | Explosive atmospheres – Part 0: General requirements. |
| IS/IEC 60079-1:2014 | Explosive atmospheres – Part 1: Equipment protection by flameproof enclosures “d”. |
| IS/IEC 60079-31:2013 | Explosive atmospheres – Part 31: Equipment Dust ignition protection by enclosure “t”. |

इस प्रकार के परीक्षण प्रमाण-पत्र उपरोक्त सूचीबद्ध मानकों में विशेष रूप से शामिल की गई अपेक्षाओं के अलावा विद्युत सुरक्षा एवं निष्पादन संबंधी अपेक्षाओं के अनुपालन को नहीं दर्शाते हैं।/ This type test certificate does not indicate compliance with electrical safety and performance requirements other than those expressly included in the standards listed above.

8. इंग्रेस सुरक्षा की मात्रा/ Degree of ingress protection: IP-66 (O ring 6mm thick gasket) as per IS IEC 60529:2001(Reaffirmed 2014)

9. परीक्षण एवं मूल्यांकन रिपोर्ट/ Test & Assessment Reports: A sample(s)/documents of the equipment listed have successfully meet the examination and test requirements as recorded in IN/CIMFR/TR23/P/H128 Dated: 12 July, 2023.

[Signature]



सीएसआईआर- केन्द्रीय खनन एवं ईंधन अनुसंधान संस्थान CSIR-Central Institute of Mining and Fuel Research



(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद/Council of Scientific & Industrial Research)
बरवा रोड, धनबाद - 826015, झारखण्ड, भारत/Barwa Road, Dhanbad - 826015, Jharkhand, India

ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF & EQUIPMENT SAFETY परीक्षण प्रमाण पत्र / TEST CERTIFICATE

प्ररूप परीक्षण प्रमाण-पत्र / TYPE EXAMINATION CERTIFICATE

Certificate No.: CMF 23 INEx 0187
ULR No.: TC23000000210F

Dated: 12/07/2023
(Issue No. 0)

SCHEDULE

11. उपकरण का विवरण/ Description of equipment: The Flameproof-Weatherproof (IP-66) Control Panel in cast aluminium Alloy LM-6 construction designated by Cat. No.: SAI/RCP/1058.

Sr. No.	Enclosure	Volume (c.c.)		Min Wall Thickness (mm)	Dimensions (mm)	Elec. Rating (Max.)	Temp Class at 40°C Amb.	Nos. & Size of Bolts/ Fasteners (mm)
		Gross	Net					
1.	Enclosure Unit No. I	90000	30000	9mm	530x780x295	630 Amps, 550V AC or upto 240V DC.	T6	Allen bolts 24 Nos. M8x 30 L
2	Enclosure Unit No. II	80000	30000	9mm	600x600x295	---do---	T6	Allen bolts 24 Nos. M8x 30 L
3	Enclosure Unit No. III	65000	25000	9mm	550x600x255	---do---	T6	Allen bolts 16 Nos. M8x 30 L
4	Enclosure Unit No. IV	50000	20000	9mm	480x480x285	---do---	T6	Allen bolts 16 Nos. M8x 30 L

12. आरेखण एवं प्रलेखन /Drawing and documents:

Sl. No.	Drg. No.	Title	Sheet	Rev.	Date
1.	SAI-1058	FLAMEPROOF - WEATHERPROOF CONTROL PANEL SUITABLE FOR USE IN GAS GR. - IIA+IIB, T - 6	1 of 1	0	13/03/2023

13. नियमित परीक्षण/ Routine Testing: A routine overpressure test in accordance with IS IEC 60079-1:2014 clause 16 shall be carried out at a pressure Result#B2 for a period of at least 10 seconds and test results must be recorded. The overpressure test shall be considered satisfactory, if no permanent deformation or damage invalidating type of protection is observed. The joints shall in no place have been permanently enlarged.

14. उपयोग के लिए विनिर्दिष्ट अवस्था/ स्थिति/शर्त / Specific conditions of use: Not Applicable

15. प्रमाण-पत्र परिवर्तन (मद 1 एवं उपर्युक्त के लिए) का विवरण/ Details of certificate changes (for issue 1 and above): Not Applicable.

16. अतिरिक्त सूचना/ Additional Information: Not Applicable

17. इतिहास/ वृत्तान्त /History: NIL

Approved for issued on behalf of CIMFR
Position

Dr. A. P. Singh
Principal Scientist & HOS

यह संस्थान सुरक्षा के हित में आवश्यक समझे जाने पर किसी भी समय इस प्रकार के परीक्षा प्रमाण पत्र की समीक्षा, संशोधन या वापस लेने का अधिकार सुरक्षित रखता है। This Institute reserves the right to review, amend or withdraw this Type Examination certificate at any time if considered necessary in the interest of safety.



सीएसआईआर - केंद्रीय खनन एवं ईंधन अनुसंधान संस्थान

CSIR - Central Institute of Mining and Fuel Research

(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद् (Council of Scientific & Industrial Research))

(अंतर्गत वैज्ञानिक तथा औद्योगिक अनुसंधान विभाग, विज्ञान और प्रौद्योगिकी मंत्रालय, भारत सरकार)

(Under the Department of Scientific & Industrial Research, Ministry of Science & Technology, Govt)

बरवा रोड, धनबाद - 826015, झारखंड, भारत / Barwa Road, Dhanbad - 826015, Jharkhand, India

(आई एस ओ 9001 प्रमाणित संस्थान / ISO 9001 Certified Institute)



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परीक्षण एवं मूल्यांकन रिपोर्ट सं.

Test & Assessment Report No. IN/CIMFR/TR23/P/ H 128

दिनांक/Dated: 12 July, 2023

पहचान संख्या/ID No. 120/23

यूएलआर सं./ULR No.: TC23000000 210F

टीसी कोड सं./TC CODE NO. FES/36/23-24

BIS QR Code No.: 100000042910

सेवा में/To

साई इंडस्ट्रीज, / M/s. SAI Industries,

प्लॉट नंबर 4709, प्लास्टिक जोन, जीआईडीसी, / Plot No. 4709, Plastic Zone, GIDC,

जिला. वलसाड, सारीगाम - 396155 (गुजरात) / Dist. Valsad, Sarigam - 396155 (Gujarat).

विषय/Subject: Type Examination Certificate along with test assessment report of Flameproof-Weatherproof (IP-66) Control Panel, in cast aluminium Alloy LM6 construction designated by Cat. No.: SAI/RCP/1058, as per IS/IEC 60079-1:2014 for use in Gas Group: IIB, EPL Gb & Db atmosphere.

- (परीक्षण रिपोर्ट/Test Report)

आपका संदर्भ आवेदन सं./ Your Ref. Appl. No.: SAI/NC/CIMFR/2023/001

दिनांक/Dated: 01/05/2023

प्रिय महोदय/Dear Sir,

आग्रह है कि कृपया आपके द्वारा भेजे गए उपरोक्त दस्तावेजों के परीक्षण मूल्यांकन रिपोर्ट के साथ संलग्न किया गया इसका टाइप परीक्षा प्रमाण-पत्र भी प्राप्त करें। Please find the enclosed Type Examination Certificate along with test assessment report of the above documents submitted by you.

इस संस्थान को सुरक्षा की दृष्टि से आवश्यकता होने पर किसी भी समय उक्त 'परीक्षण रिपोर्ट' की समीक्षा, संशोधन करने या उसे वापस लेने का अधिकार प्राप्त है। This Institute reserves the right to review, amend or withdraw this 'Test Report' at any time, if considered necessary in the interest of safety.

इस परीक्षण कार्य के लिए लागू सेवा जीएसटी सहित रु. 1,26,354/- के शुल्क को आपके द्वारा किए गए अग्रिम जमा के साथ समायोजित किया गया है। Charges of Rs. 1,26,354/- (Rupees One Lakh Twenty Six Thousand Three Hundred Fifty Four Only) including applicable GST involved towards the testing have been adjusted against the advance deposit made by you.

कृपया इस पत्र की प्राप्ति की तारीख से 90 दिनों के भीतर नमूना एकत्र करने की व्यवस्था करें, ऐसा न करने पर सीआईएमएफआर आपको बिना किसी सूचना के सार्वजनिक नीलामी द्वारा नमूनों का निपटान करेगा। /Kindly arrange to collect the sample within 90 days from the date of receipt of this letter failing which CIMFR would dispose of the samples by public auction without any further NOTICE to you.

कृपया प्राप्ति सूचना भेजें। Kindly acknowledge receipt.

सधन्यवाद/Thanking you.

भवदीय/Yours faithfully,

जितेंद्र कुमार सिंह
12/07/2023

(जे. के. सिंह/J.K. SINGH)

अनुभाग प्रमुख, परीक्षण प्रकोष्ठ
HOS, TESTING CELL

संलग्न/Encl: 1. परीक्षण मूल्यांकन रिपोर्ट/Test & assessment report in Four Copies

2. टाइप परीक्षा प्रमाण-पत्र/Type examination certificate

प्रति/Copy to: 1. प्रमुख, ज्वाला-सह एवं उपस्कर सुरक्षा/Head, Flameproof & equipment Safety

2. बिल अनुभाग/Bill section

HQ : Dhanbad - ☎: +91-326-2296023, 2296006 (O) 📠: +91-326-2296025 EPABX : +91-326-2296012/6013/6027/6028

E-mail : director@cimfr.nic.in/drpsingh@cimfr.nic.in Digwadih Campus : ☎ +91-326-2381111

Research Centres : Bilaspur : ☎ +91-775-2271450, Nagpur : Unit - I - ☎ +91-712-2510604, Unit - II - ☎ 2510390

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सीएसआईआर- केन्द्रीय खनन एवं ईंधन अनुसंधान संस्थान CSIR-Central Institute of Mining and Fuel Research



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ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF & EQUIPMENT SAFETY

परीक्षण रिपोर्ट / TEST REPORT

मूल प्रतिलिपि
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FORMAT NO.: (CIMFR: DQM: FLP02: F-01:REV-01)

(Flameproof & Equipment Safety)

Equipment ID No.: 120/23

ULR No.: TC23000000210F

Code No.: FES/36/23-24

Test & Assessment Report No.: IN/CIMFR/TR23/P/ H128 Dated: 12 July, 2023
BIS QR CODE: 100000042910
Application Ref. No.: SAI/NC/CIMFR/2023/001 Dated: 01/05/2023

- Applicant : M/s. SAI Industries,
Plot No. 4709, Plastic Zone, GIDC,
Dist. Valsad, Sarigam - 396155 (Gujarat).
- Manufacturer : Same as Above.
- Equipment : Flameproof-Weatherproof (IP-66) Control Panel.
- Designated by : Cat. No.: SAI/RCP/1058.
- Type of protection : Ex db for Gas Group: IIB, EPL Gb & Db.
- Electrical ratings : Rated 630A, 550VAC or 240VDC (max.) (Tested on manufacturer site)
- Temperature Class : T6 @ 40°C amb
- Degree of Ingress Protection : IP-66 (O ring 6mm thick gasket)
- Material of Construction : Cast Aluminium Alloy LM- 6
- Description of the equipment: Flameproof-Weatherproof (IP-66) Control Panel.

Sr. No.	Enclosure	Volume (c.c.)		Min Wall Thickness (mm)	Dimensions (mm)	Elec. Rating (Max.)	Temp Class at 40°C Amb.	Nos. & Size of Bolts/ Fasteners (mm)
		Gross	Net					
1.	Enclosure Unit No. I	90000	30000	9mm	530x780x295	630 Amps, 550V AC or upto 240V DC.	T6	Allen bolts 24 Nos. M8x 30 L
2	Enclosure Unit No. II	80000	30000	9mm	600x600x295	---do---	T6	Allen bolts 24 Nos. M8x 30 L
3	Enclosure Unit No. III	65000	25000	9mm	550x600x255	---do---	T6	Allen bolts 16 Nos. M8x 30 L
4	Enclosure Unit No. IV	50000	20000	9mm	480x480x285	---do---	T6	Allen bolts 16 Nos. M8x 30 L

B

Sl. No.	Glass Details (Tested Sealing material- Epoxy Resin Compound, COT: -20°C to 220°C Heat & Impact Resistance Toughened glass.	Max Nos. of aperture*
1.	Enclosure Unit No. I 1 No. Glass size = 300x200mm, window, Shape = Rectangular, cemented path ≥ 15mm. thickness = 12mm	R.S. = 18 Nos., P.B. = 22 Nos., IL= 18 Nos. OR suitable combination as per requirements = (Max. Nos. of apertures on caver – 58 Nos.)
2.	Enclosure Unit No. II 1 No. Glass size = 300x430mm, window, Shape = Rectangular, cemented path ≥ 15mm. thickness = 15mm	R.S. = 7 Nos., P.B. = 7 Nos., IL= 8 Nos. OR combination as per requirements = (Max. Nos. of apertures on caver – 22 Nos.)
3.	Enclosure Unit No. III 1 Nos. Glass size = 300x200mm, window, Shape = Rectangular, cemented path ≥ 15mm. thickness = 10mm	R.S. = 9 Nos., P.B. = 9 Nos., IL= 8 Nos. OR combination as per requirements = (Max. Nos. of apertures on caver – 26 Nos.)



सीएसआईआर- केन्द्रीय खनन एवं ईंधन अनुसंधान संस्थान CSIR-Central Institute of Mining and Fuel Research

(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद) / (Council of Scientific & Industrial Research)

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ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF EQUIPMENT SAFETY

परीक्षण रिपोर्ट / TEST REPORT

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FORMAT NO.: (CIMFR: DQM: FLP02: F-01:REV-01)

(Flameproof & Equipment Safety)

Equipment ID No.: 120/23

ULR No.: TC23000000210F

Code No.: FES/36/23-24

4.	Enclosure Unit No. IV 2 Nos. Glass size = 105x105mm, window, Shape = Square, cemented path $\geq$ 15mm. thickness = 10mm	R.S. = 8 Nos., P.B. = 8 Nos., IL = 8 Nos. OR combination as per requirements = (Max. Nos. of apertures on cover - 24 Nos.)
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11. Nature of Flameproof Joint: Type of joints and gaps (Spigot joint, Cylindrical, Flange Joint & Threaded Joint)

Sl. No.	Description (Between Flamepath and location)	Type of joint	Min. Axial length Flamepath (mm)		Max dia. Clearance / gap in (mm) / No. of thread	
			Req.	Spec.	Req.	Spec.
1.	Encl. body & Cover	Flange	l = 9 L = 25	l = 13 L = 27	0.15	0.15
2.	Encl. body & Brass bush	Cylindrical	12.5	15	0.15	Press fit
3.	Brass bush & operating rod (spindle)	Cylindrical	12.5	15	0.15	0.15
4.	Encl. body & indicating lamp	Threaded	8	13	5T	8T(1.5P)
5.	Brass bush & Enclosure body	Threaded	8	13	5T	8T(1.5P)
6.	Enclosure body & coupling check nut	Threaded	8	13	5T	8T(1.5P)
7.	Cable entry and body of enclosure	Threaded	8	13	5T	8T(1.5P)

Max. No. & size of cable entries are as given below chart: The maximum number of cable entries and sizes allowed with certified and approved FLP/WP double compression type cable gland / blanking plug / breather drainer can be used along with these entries. The axial length and threads are provided as per mention above table.

Max. No. & size of cable entries: The maximum number of following certified and approved FLP/WP double compression type cable gland/blanking plug/breather drainer can be used as following.

Nos. of Cable entries provided in test sample from each side								
TYPE NO.	M12/16/20 or ½" NPT	M25 or ¾" NPT	M32 or 1" NPT	M40 or 1.1/4" NPT	M50 or 1.1/2" NPT or 2" NPT	M63 or 2.1/2" NPT	M75 or 3" NPT	M82/90/100 or 3.5" NPT
Enclosure Unit No. I	51	40	30	20	10	8	6	4
Enclosure Unit No. II	44	35	25	17	8	7	6	4
Enclosure Unit No. III	44	35	25	17	8	7	6	4
Enclosure Unit No. IV	29	24	20	15	8	6	5	3

12. Name plate, Marking & Warning inscription: The name-cum-warning inscription plate and rating plate made of Brass / SS is permanently fixed on the Cover of enclosure by hammer driven rivets / screwed leaving min. 3mm material thickness below the rivets / screw. The warning inscription shall read as "DO NOT OPEN WHEN ENERGIZED".

13. Drawings: These unit are designed and constructed as flameproof and weatherproof in accordance with the following drawing conforming to the requirements of IS/IEC 60079-0:2017, IS/IEC 60079-1: 2014 and IS/IEC 60529-2001 (Reaffirmed 2014).

Sl. No.	Drg. No.	Title	Sheet	Rev.	Date
1.	SAI-1058	FLAMEPROOF - WEATHERPROOF CONTROL PANEL SUITABLE FOR USE IN GAS GR. - IIA+IIB, T - 6	1 of 1	0	13/03/2023

12-7-23



सीएसआईआर- केन्द्रीय खनन एवं ईंधन अनुसंधान संस्थान
CSIR-Central Institute of Mining and Fuel Research

[Continuation Sheet]



(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्) / (Council of Scientific & Industrial Research)
बरवा रोड, धनबाद - 826015, झारखण्ड, भारत / Barwa Road, Dhanbad - 826015, Jharkhand, India

ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF & EQUIPMENT SAFETY

परीक्षण रिपोर्ट / TEST REPORT

मूल प्रति
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FORMAT NO.: (CIMFR: DQM: FLP02: F-01:REV-01)
(Flameproof & Equipment Safety)

Equipment ID No.: 120/23

ULR No.: TC23000000210F

Code No.: FES/36/23-24

14. Any other relevant information:

- As per Clause 4.3 Note - 3 of IS/IEC 60079-0:2017 the equipment mark IIB is suitable for application requiring Gas Group IIA or IIB equipment. And as per Clause 4.4 Note of IS/IEC 60079-0:2017 the equipment mark IIIC is suitable for application requiring Dust Group IIIA or IIIB equipment.
- The material of construction of the said panel can be in Cast Iron or SS or Brass or M.S Fabricated instead of cast Aluminum Alloy LM-6.
- Equal and equivalent NPT threads of medium class fit conforming to requirements of ANSI/ASME B1.20.1 may be used instead of metric threads conforming to ISO 965-1 & 965-3 as per Clause No. 5.3 & 13 of IS/IEC 60079-1:2014.
- The above equipment may also be used in Dust Group-IIIC, EPL Db & Dc area as per IS/IEC 60079-31: 2013.
- The extensible type control panel board enclosures is suitable for application as control panel, lighting or power control panel, distribution board, control station, motor control centre, compressor control panel, process or hoist / crane control panel, annunciator or instrumentation panel, soot blower control panel, motor starter, weighing scale panel etc.
- Following electrical equipment's can be housed / fitted inside the enclosures.
 - Dol / star delta / reversing starter suitable for upto 400 HP or fuse switch unit/ reversing switch / isolator switch / MCCB / MCB / ELCB / ELCB+MCB / RCCB / RCBO / Switch Socket and Plug Unit / Fuses upto 630 amps.
 - Push button control stations, indicating lamp, relays, PLC, POT, Zenner barrier, relay cards, time switch, timers, CT / PT, Ammeter, Voltmeter, / Control Switches, hooter / analog or digital instruments or any other control equipment / instruments etc. & electrical & electronic components, transformers upto maximum 10000VA, ignition transformers upto maximum 20KVA, Ignitor circuit, copper or aluminium Busbar, VFD, power capacitors / overload relays upto 630A.
 - Temperature Indicators, Temperature Controllers, Digital / Analog Instruments / Controllers / Timer Or Programmer Or Level Controllers etc....
- The enclosure may be provided with glass window maximum aperture size mentioned in Table B. The flameproof features and max. Size of glass windows will be as shown in the drawing. Behind glass window any electrical/electronic instruments / indicating lamps etc. Can be fitted.
- The enclosures may be used as instrumentation box / terminal enclosures / junction box / bus bar enclosure. The number of cable entries can be as per "table - Clause No.11" - (suitable combination of cable entries as per requirements).

[Signature]
12.7.23



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[Continuation Sheet]



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बरवा रोड, धनबाद - 826015, झारखण्ड, भारत / Barwa Road, Dhanbad - 826015, Jharkhand, India

ज्वालासह एवं उपकरण सुरक्षा / FLAMEPROOF & EQUIPMENT SAFETY

परीक्षण रिपोर्ट / TEST REPORT

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FORMAT NO.: (CIMFR: DQM: FLP02: F-01:REV-01)

(Flameproof & Equipment Safety)

Equipment ID No.: 120/23

ULR No.: TC23000000210F

Code No.: FES/36/23-24

15. Declaration by the Applicant/Manufacturer: (i) As to standards with which the equipment complies in respect of:
- Electrical equipment for Potentially Explosive atmosphere – General requirements: IS/IEC 60079-0: 2017
 - As to flameproof protection IS/IEC 60079-1: 2014
 - Ingress protection as per IS/IEC 60529: 2001 (Reaffirmed 2014)
 - Dust ignition protection by enclosure 't' IS/IEC 60079-31:2013

(ii) Composition of the cast aluminum alloy declared by the manufacturer: Composition of cast aluminum alloy purported to be forming the material of construction of the enclosure has been declared by the manufacturer in the drawing having 7.5% in total of magnesium titanium and zirconium.

However no sample of the alloy was drawn from the prototype enclosure for verifying its chemical composition declared by the manufacturer.

16. Documents/Samples Submitted: (i) Application form (ii) Drawings (iii) Test Sample

17. Condition of Sample received: Good

Note: CIMFR has however not checked and tested the compliance of the equipment to any standard other than the above standards.

SCOPE OF THE TEST CERTIFICATE

The Test Certificate issued by CIMFR certifies that the equipment has been found to comply with the definition of Flameproof-Weatherproof equipment contained in the relevant Standard specifications. They do not vouch for the quality of the equipment in any other respect. The present test report pertains to the sample received only.

This Institute reserves the right to review, amend or withdraw this Test Report at any time if considered necessary in the interest of safety.

REPORT OF TEST

Date of Test: 22/06/2023

Test Equipment Used: (1) Computer, Kistler Piezo Electric Transducer and Charge Amplifier for conducting Explosion Pressure Tests, (2) Overpressure Test Setup.

Result #A: Tests as per IS/IEC 60079-0: 2017:

Type Tests			
Clause	Tests	Remarks	Results (Complies, P- Pass, NA-Not Applicable, F-Fail) References
1	Scope		Complies
2	Normative references		Complies
3	Terms and definitions		Complies
4	Equipment grouping	Gas Group: IIB, EPL Gb/Db	Complies
5	Temperature		
5.1.1	Ambient Temperatures	- 20 ⁰ C to +50 ⁰ C Amb.	Complies
6	Requirements for all electrical equipment		Complies
6.6	Electromagnetic and ultrasonic energy radiating equipment.		NA
7	Non-metallic enclosures and non-metallic parts of enclosures		NA
7.1.2.4	Material used for cementing / Sealing	Tested sealing material shall be used	Complies
7.2	Thermal Endurance		Complies

12.7.23



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8	Metallic Enclosure and metallic part of Enclosures	Cast Aluminium alloy construction	Complies
9	Fasteners	450N/mm ²	Complies
10	Interlocking devices	Check nut	Complies
11	Bushings		Complies
12	Reserved for future use		NA
13	Ex components		NA
14	Connection facilities		Complies
15	Connection facilities for earthing and bonding conductors		Complies
16	Entries into enclosures		Complies
17	Supplementary requirements for electric machines		NA
18	Supplementary requirements for switchgear		NA
19	Reserved for future use		NA
20	Supplementary requirements for external plugs and socket outlets and connectors for field wiring connection		NA
21	Supplementary requirements for luminaries		NA
22	Supplementary requirements for cap lights and hand lights		NA
23	Equipment incorporating cells and batteries		NA
24	Documentation		Complies
25	Compliance of prototype or sample with documents		Complies
26	Type tests		Complies
26.3	Tests in explosive test mixtures		
26.3	Tests for Flameproof (Ex 'd') protection	Verification for Compliance to Ex 'd' requirements as per IS/IEC 60079-1: 2014	P/As per Result #B
26.4	Test of enclosures		Complies
26.4.2	Resistance to Impact		
26.4.3/26.4.4	Drop test	(Not a handheld equipment or portable device) NA	
26.4.5	Degree of protection (IP) by Enclosures		
26.4.5	Test of IP of equipment	IP-66 (O ring gasket 6mm thick)	
26.5	Thermal tests		
26.5.1	Temperature measurement	T6 at 40°C amb.(110% rated current) Tested on manufacturer site	
26.5.2	Thermal shock test		Complies
26.6	Torque Test for bushings		
26.6	Torque test for bushings		NA
26.7	Non-metallic enclosures or non-metallic parts of enclosures		
26.8	Thermal endurance to heat	Tested sealing material shall be used	
26.9	Thermal endurance to cold		
26.10	Resistance to UV light		NA
26.11	Resistance to chemical agents for Group I Equipment	-----	NA
26.12	Earth continuity test in non-metallic enclosure		NA
26.13	Surface resistance test of parts of enclosures of Non-metallic materials	-----	NA
26.14	Measurement of capacitance		NA
26.15	Verification of ratings of ventilating fans		NA
26.16	Alternative qualification of elastomeric sealing O-rings	Neoprene / Silicone	Complies
27	Routine tests	Manufacturer's responsibility: Compliance to prototype of the product and physical measurement of gap and flamepath as per test report. Overpressure test as per Result#B2 for every sample.	
28	Manufacturer's responsibility	Conformity of the equipment to the applicable standards	Complies
29	Marking	(Ex db IIB T6 Gb) (Ex tb/tc IIIC T85°C Db/Dc)	

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Code No.: FES/36/23-24

30	Instructions	Instruction related to use, installation, maintenance, adjustment to be provided by the manufacturer along with the equipment.
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Result #B: Tests in explosive mixtures: Tests for type of protection Ex 'd' as per IS/IEC 60079-1:2014.

Type Tests			
Clause No.	Description and relevant tests	Remarks	Results (Complies, P- Pass, NA-Not Applicable, F-Fail) References
1	Scope		Complies
2	Normative references		Complies
3	Terms and definitions		Complies
4	Level of protection (Equipment Protection Level, EPL)	Gas Group IIB, EPL-Gb/Db	Complies
5	Flameproof joints	Threaded & cylindrical & Flange	Complies
6	Sealed joints		
6.1	Cemented joints		Complies
6.1.2	Mechanical strength		Complies
7	Operating rods		Complies
8	Supplementary requirements for shafts and bearings		NA
9	Light transmitting parts		NA
10	Breathing and draining devices which form part of a flameproof enclosure		NA
11	Fasteners and openings	450N/mm ²	Complies
12	Materials	Cast Aluminium Alloy construction	Complies
13	Entries for flameproof enclosures		Complies
14	Verification and tests		Complies
15	Type Tests		
15.1	General		Complies
15.2	Tests of ability of the enclosure to withstand pressure		
15.2.1	General		Complies
15.2.2	Determination of Explosion Pressure (Reference Pressure)		P/As per Result #B1
15.2.3	Overpressure Test		P/As per Result #B2
15.3	Test for non-transmission of an internal ignition		
15.3.1	General		Complies
15.3.2	Electrical equipment of Groups I, IIA & IIB		P/As per Result #B3
15.3.3	Electrical equipment of Group IIC		NA
15.4	Tests of flameproof enclosures with breathing and draining devices		
15.4.1	General		NA
16	Routine tests	To be conducted by the manufacturer as per test Result #B2 for every sample.	
17	Switchgear for Group I		NA
18	Lamp holders and lamp caps		NA
19	Non-metallic enclosures and non-metallic parts of enclosures		
19.1	General		NA
19.2	Resistance to tracking and creepage distances on internal surfaces of the enclosure walls		NA
19.3	Requirements for type test		Complies
19.4	Test of erosion by flame		NA
20	Marking	(Ex db IIB T6 Gb) (Ex tb/tc IIIC T85°C Db/Dc)	
20.2	Caution and warning markings	"DO NOT OPEN WHEN ENERGIZED"	
21	Instruction	The manufacturer shall supply instruction manual of the equipment complying IS/IEC 60079-0:2017.	

[Signature]
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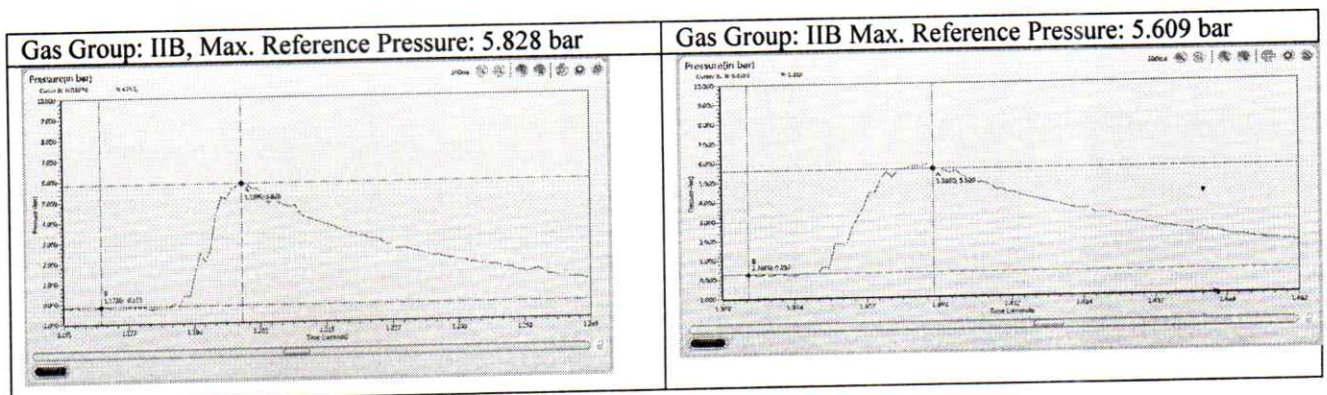
Result # B1: Determination of Explosion pressure (Reference Pressure Test) *

Gas mixture: Test with 8% Ethylene in air				
No. of Tests: 3 nos. test with Single enclosure. Test Condition: Enclosure fitted with component and clause 15.2.2.2				
Type of Test	Gas Group		Gas Mixture % in Air	No. of Tests
Preliminary Test (reference pressure)	IIB		8% Ethylene in air	Three
Test Ref. No.	Ignition	Gauge Position	Max. Pressure in Bar	Remark
Enclosure Unit No. I				
PPM/1202301	Body	Body	4.875	No evidence of distress
PPM/1202302	Body	Body	5.169	-do-
PPM/1202303	Body	Body	5.828	-do-
Enclosure Unit No. II				
PPM/1202304	Body	Body	5.535	No evidence of distress
PPM/1202305	Body	Body	5.535	-do-
PPM/1202306	Body	Body	5.609	-do-
Enclosure Unit No. III				
PPM/1202307	Body	Body	5.462	No evidence of distress
PPM/1202308	Body	Body	5.535	-do-
PPM/1202309	Body	Body	5.609	-do-
Enclosure Unit No. IV				
PPM/1202310	Body	Body	5.242	No evidence of distress
PPM/1202311	Body	Body	5.242	-do-
PPM/1202312	Body	Body	5.169	-do-

*Results shown for the pressure curve (enclosed) are the highest recorded value obtained in the test.

Result #B2: Overpressure Test (Static Method): Over pressure test is conducted as per Cl. 15.2.3.2 of IS/IEC 60079-1: 2014.

Enclosure	Test reference nos.	Over pressure (kg/cm ²) maintained for 10 sec.	Remark
Enclosure Unit No. I	OPM/1202313	9 kg/cm ²	No damage or deformation
Enclosure Unit No. II	OPM/1202314	8.5 kg/cm ²	-do-
Enclosure Unit No. III	OPM/1202315	8.5 kg/cm ²	-do-
Enclosure Unit No. IV	OPM/1202316	7.5 kg/cm ²	-do-





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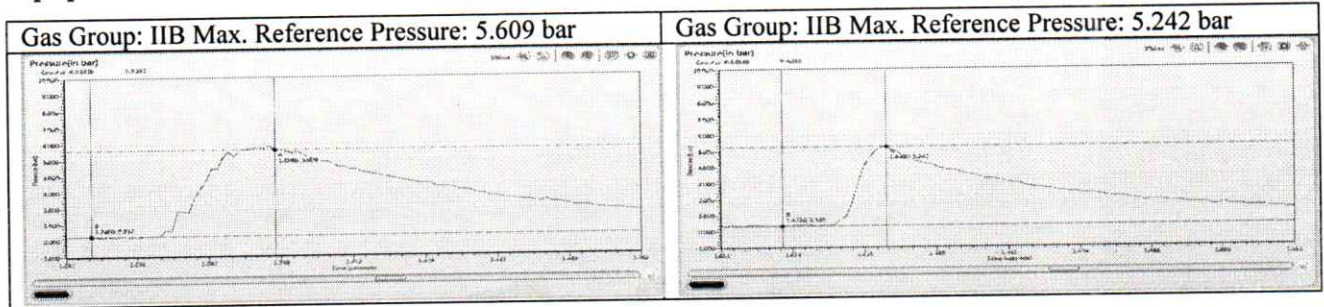
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Result #B3: Test for Non transmission of internal ignition:

Gas Mixture:	Test with 37% Hydrogen in air for Gas Gr. IIB				
No. of Tests:	5 nos. tests with Single enclosure. Test Condition: Clause 15.3.2.1 (No pre compression required)				
Name of Enclosure	Test Ref. Nos.	Gas Group	Gas Mixture % in Air	No. of Test	Result
Enclosure Unit No. I	EIT/1202317 to EIT/1202321	IIB	37% H ₂	Five	Pass
Enclosure Unit No. II	EIT/1202322 to EIT/1202326	IIB	37% H ₂	Five	Pass
Enclosure Unit No. III	EIT/1202327 to EIT/1202331	IIB	37% H ₂	Five	Pass
Enclosure Unit No. IV	EIT/1202332 to EIT/1202336	IIB	37% H ₂	Five	Pass

Conclusion: The FLP-WP (IP-66) Control Panel, designated by Cat. No.: SAI/RCP/1058 in Cast Aluminium Alloy LM 6 construction, meets the test requirements for flameproof-ness as per IS/IEC 60079-1:2014 and general requirements as per IS/IEC 60079-0:2017 and dust ignition protection as per IS/IEC 60079-31: 2013 is deemed to be suitable for use in Gas Group: IIB, EPL Gb & Db atmosphere.

Reported By

Checked & Approved By

12-7-23

(R. K. Mishra)
Sr. Technical Officer (2)

Dated: 12th July, 2023

Flameproof & Equipment Safety

CSIR - Central Institute of Mining & Fuel Research,
Barwa Road, Dhanbad - 826 015, (Jharkhand) India

12/7/23

(A. P. Singh)
Pr. Scientist & HOS